

Dependency Preserving Decomposition

B alone not sk

R(A, B, C)

C alone not sk

check / derive FDS,

1	1	1
2	1	2
3	2	1
4	2	2

 $A \rightarrow B, A \rightarrow C, AB \rightarrow C,$ $\uparrow$ $AC \rightarrow B$ $A \rightarrow BC$

unique

 $BC \rightarrow A$

must be

as key

FD { $A \rightarrow B, A \rightarrow C, A \rightarrow BC$ } $C \rightarrow B \times, C \rightarrow A \times$ $B \rightarrow C \times \in B \rightarrow A \times$ $\Downarrow$

Decompose

 R_1

A B

1 1

2 1

3 2

4 2

FD: { $A \rightarrow B$ } $B \rightarrow A \times$ R_2

B C

1 1

1 2

2 1

2 2

{ $B \rightarrow C \times$ $C \rightarrow B \times$ $FD_2 = \{ \text{null} \}$ $FD_1 \cup FD_2 - \text{null} = FD$

Dependency

Preserving Rule

By combining

(R1) (R2)

 $FD_1 \cup FD_2 = FD$

we must have all

FD of

Base Relⁿ

only one

FD remain,

Remaining FD loss

 $A \rightarrow B = FD \{ A \rightarrow B, A \rightarrow C, A \rightarrow BC \}$ This Decompose of R_1 & R_2 fails & dependency not preserved.

Rule = Dependency must not be loss

It should be preserved

Ex $\Rightarrow R(ABED E)$

check decomposition is lossless or not?

FD: $\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$

decomp $\Rightarrow R_1(A, B, C), R_2(C, D, E)$

soln:

$R_1(A, B, C)$

$R_2(C, D, E)$

$FD_1 = \{A \rightarrow B, B \rightarrow C\}$ $C \rightarrow D$ wrong approach

check decomposition steps

① find closure of each attributes of R_1 & find new FD

$A^+ \{A, B, C\} \Rightarrow A \rightarrow BC$
 $B^+ \{B, C, D, A\} \Rightarrow B \rightarrow CA$
 $C^+ \{C, D, A, B\} \Rightarrow C \rightarrow AB$

$\left. \begin{array}{l} A^+ \\ B^+ \end{array} \right\} \begin{array}{l} D \text{ not part of } R_1 \\ B, C \text{ are repeating} \end{array}$

② find closure of R_2 & find new FD

$C^+ = \{C, D, A, B\} \Rightarrow C \rightarrow D$
 $D^+ \{D, A, B, C\} \Rightarrow D \rightarrow C$
 $E^+ \{E\} \Rightarrow \text{not derive anything}$

check, Base FD = $(FD_1 \cup FD_2)$

All Base FDs, directly or indirectly part of union of FD_1 & FD_2

$R_1: FD_1 = \{A \rightarrow BC, B \rightarrow CA, C \rightarrow AB\}$

$R_2: FD_2 = \{C \rightarrow D, D \rightarrow C\}$

Dependency preserving.
 By union $FD_1 \cup FD_2$ is
 $\{A \rightarrow BC, B \rightarrow CA, C \rightarrow AB, C \rightarrow D, D \rightarrow C\}$

Decomposition must be lossless

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Ex → $R(AB CDE)$

FD: $\{A \rightarrow BCD, B \rightarrow AE, BC \rightarrow AFD, D \rightarrow E, C \rightarrow DE\}$

$R_1(A, B)$

$R_2(B, C)$

$R_3(C, D, E)$

check decomposition is ~~lossless~~ or ~~lossy~~?
dependency preservation

Soln

$R_1(A, B)$

$R_2(B, C)$

$A^+ = \{A, B, C, D, E\}$

$B^+ = \{B, A, E, C, D\}$

$(AB)^+ = \{A, B, C, D, E\}$ $A \rightarrow B$

$B \rightarrow C$

$BC^+ = \{B, C, A, E, D\}$

Repeating
A & B

$B^+ = \{B, A, E, C, D\}$

$C^+ = \{C, D, E\}$
No FD possible

as B & C

$B \rightarrow A$

$R_3(C, D, E)$

$CD^+ = \{C, D, E\}$

$C^+ = \{C, D, E\}$

$C \rightarrow DE$

Repeating

$D^+ = \{D, E\}$

$D \rightarrow E$

$C \rightarrow E$

$E^+ = \{E\}$

$D \rightarrow E$

$FD_1 \cup FD_2 \cup FD_3 = FD$

(G)

$\{A \rightarrow B, B \rightarrow A, B \rightarrow C, C \rightarrow DE, D \rightarrow E\} = FD$

$A^+ = \{A, B, C, D, E\}$

$B^+ = \{B, A, C, D, E\}$

$BC^+ = \{B, C, D, E, A\}$

By checking all FD of
 $R(AB CDE)$ with
Union of FD_1, FD_2, FD_3

$\therefore G \subseteq F$

Relation's Decomposition is

lossless. Dependency preserved

Dependency Preserving Criteria

$$\underbrace{F_1 \cup F_2}_{G} = F$$

G must cover F

$$G \supseteq F$$

F cover G

$$\cancel{G \supseteq F} \quad G \subseteq F$$

Each members of G must be present in F & vice versa
(FDs)

Next
step of
Dependency
preservation



After Union Op

To check Dependency preservation At last

→ Take the closure of LHS of F 's $\forall$ Fd.

Ex $\Rightarrow R(A, B, C, D)$

FD: $\{A \rightarrow B, C \rightarrow D\}$

$R_1(A, C)$

$R_2(B, D)$

check given Relation's decomposition is ~~lossy~~ or ~~lossy~~
Dependency Preserve or not

Soln $\Rightarrow$

$R_1(A, C)$

$R_2(B, D)$

$A^+ = \{A, B\}$

$B^+ = \{B\}$

$\therefore$ No FD $A \rightarrow B$ B is not part of

$D^+ = \{D\}$

} not derived any FDs

$C^+ = \{A, C, D\}$ R_1

$\therefore$ No FD $C \rightarrow D$ D is not part of

$(AC)^+ = \{A, B, C, D\}$ R_2

FD₂ = Does Not Exist

↑
total = Repeating

FD₁ = $\{A \rightarrow B, C \rightarrow D\}$
DNE

$\underbrace{FD_1 \cup FD_2}_{\rightarrow G_1} = \underbrace{FD}_F$

No new FD generated.

~~$\{A \rightarrow B, C \rightarrow D\}$~~ = ~~$\{A \rightarrow B, C \rightarrow D\}$~~

DNE = FD

$G_1 \neq FD$

It is lossy decomposition

& no dependency is preserving

Property

$R_1 \cap R_2 = \emptyset$

Nothing common
(Attribute)
between R_1 & R_2
 $\therefore$ Lossy Decomposition
& no FD Preserve

Conclusion

Condition for Dependency Preserving

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$G \equiv F \checkmark \rightarrow$ Dependency Preserve

$G \subset F \quad \times DP$

$F \subset G$ not possible

$F \subset G$ never has more / Extra FD's
than F .

Note $\Rightarrow$ To "ensure" Dependency preserve, it is
important to take closure of All
"Attribute" at least in order to
form desired FD's.

upto 3NF dependency Preservation is Possible

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Ex $\Rightarrow$ $R(A, B, C, D, E)$ check for decomposition is dependency Preservation or not?

FD: $\{A \rightarrow B, C, D, E, BC \rightarrow A, E, D \rightarrow E\}$

$R_1(A, B)$, $R_2(B, C)$, $R_3(C, D, E)$

Soln:

$R_1(A, B)$	$R_2(B, C)$	$R_3(C, D, E)$
$A^+ = \{A, B, C, D, E\}$	$B^+ = \{B\}$	$C^+ = \{C\}$
$A \rightarrow B$	$C^+ = \{C\}$	$D^+ = \{D, E\}$
$B^+ = \{B\}$	$(BC)^+ = \{B, C, A, E, D\}$	$D \rightarrow E$
New FD ₁ = $\{A \rightarrow B\}$	No new FD ₂ possible	$E^+ = \{E\}$
		new FD ₃ = $\{D \rightarrow E\}$

$$FD_1 \cup FD_2 \cup FD_3 = FD$$

$$\{A \rightarrow B, D \rightarrow E\} \neq \{A \rightarrow B, C, D, E, BC \rightarrow A, E, D \rightarrow E\}$$

As one attribute is missing 'C'

FD lost in decomposition = $A \rightarrow B, C, D, E$ & $BC \rightarrow A, E, D$

$\therefore$ not dependency preserving

Ex: $\Rightarrow R(A, B, C, D, E)$ check for decomposition

FD: $\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$ is ~~lossy~~ or not?
dependency preserving

$R_1(A, B)$

$R_2(B, C), R_3(C, D)$

Soln:

$R_1(A, B)$	$R_2(B, C)$	$R_3(C, D)$
$AT = \{A, B, C, D\}$ $A \rightarrow B$	$BT = \{B, C, D, A\}$ $B \rightarrow C$	$CT = \{C, D, A, B\}$ $C \rightarrow D$
$BT = \{B, C, D, A\}$ $B \rightarrow A$	$CT = \{C, D, A, B\}$ $C \rightarrow B$	$DT = \{D, A, B, C\}$ $D \rightarrow C$
$ABT = \text{Repeating}$	$BCT = \text{Repeating}$	$DCt = \{ \text{Trivial} = \text{Repeating} \}$
$FD_1 = \{A \rightarrow B, B \rightarrow A\}$	$FD_2 = \{B \rightarrow C, C \rightarrow B\}$	$FD_3 = \{C \rightarrow D, D \rightarrow C\}$

$$\therefore \underbrace{FD_1 \cup FD_2 \cup FD_3}_G = \underbrace{FD}_F$$

$$\{A \rightarrow B, B \rightarrow A, B \rightarrow C, C \rightarrow B, C \rightarrow D, D \rightarrow C\} = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, \textcircled{D \rightarrow A}\}$$

$$AT = \{A, B, C, D\}$$

$$BT = \{A, B, C, D\}$$

$$CT = \{A, B, C, D\}$$

$$DT = \{A, B, C, D\}$$

$$(D \rightarrow A)$$

$$G \equiv F$$

is Not directly visible here
but derive in AT closure
 $\therefore$ It is ~~lossy~~ decomposition
dependency preserving
satisfy

"Lossless Join Decomposition"

- if Relation, Reconstruct from sub relation then it is lossless join
- check with the Natural join ($\bowtie$)

Natural join is improved version of Cartesian Product ($\times$)

Ex: $\Rightarrow$

$R_1(A, B)$

$R_2(B, C)$

1	1	1	1
2	1	1	2
3	2	2	1

(Cartesian Product)

$R_1 \times R_2$

$R_1 \bowtie R_2$ (Natural Join)

	$R_1 \times R_2$				$R_1 \bowtie R_2$ (Natural Join)				
	A	B	B	C	A	B	B	C	
9 tuple	1	1	1	1	1	1	1	1	5 tuple
	1	1	1	2	1	1	1	2	
	1	1	2	1					
	2	1	1	1	2	1	1	1	
	2	1	1	2	2	1	1	2	
	2	1	2	1					5 tuple
	3	2	1	1					
	3	2	1	2					
	3	2	2	1	3	2	2	1	

$\text{num(Attribute)} = 4$
(column) = 4

$\text{num(row/tuple)} = 9$

Attribute not need to write again as duplicate.

Natural Join $\equiv$ Cartesian Product

↳ if No Common Attribute is Present.

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⇒ If No Common Attributes present in Relation 1 & Relation 2,

And we perform Natural Join,

then Result is Cartesian product only

Example: ⇒

$$\left. \begin{array}{l} \text{Natural} \\ \text{Join} \end{array} \right\} = \text{Cartesian Product} \left\{ \begin{array}{l} \text{when NO Common} \\ \text{Attributes exists} \end{array} \right.$$

⇒ Natural Join is only Binary operation,

we can not perform multiple joins at the same time,

it has to do multiple join one by one as.

a join of 2 relation at a time

check for lossless join or not?

Ex $\Rightarrow$

$R(A, B, C)$

$R_1(A, B)$

$R_2(B, C)$

1 1 1

1 1

1 1

2 1 2

2 1

1 2

3 2 1

3 2

2 1

4 3 2

4 3

3 2

Solⁿ $\Rightarrow$

Cartesian Product

Natural Join

$R_1 \times R_2$

$R_1 \bowtie R_2$

A B B C

A B B C

(Attributes)

1 1 1 1 $\rightarrow$

1 1 1 1

num(column)

1 1 1 2 $\rightarrow$

1 1 1 2

= 4

1 1 2 1 $\times$

num(row)

1 1 3 2 $\times$

tuples

num(row)

2 1 1 2 $\rightarrow$

2 1 1 1

= 6

= tuples

2 1 2 2 $\rightarrow$

2 1 1 2

= 16

2 1 2 1 $\times$

2 1 3 2 $\times$

3 2 1 1 $\times$

num(column)

3 2 1 2 $\times$

Attributes

Because of

3 2 2 1 $\rightarrow$

3 2 2 1

= 3

Redundancy

3 2 3 2 $\times$

in Relation

4 3 1 1 $\times$

$\therefore$ Extra tuples

4 3 1 2 $\times$

generated

4 3 2 1 $\times$

4 3 3 2 $\rightarrow$

4 3 3 2

4 3 2

$R_1 \bowtie R_2$

$R(ABC)$

A B C

1 1 1

1 1 2

2 1 1

2 1 2

3 2 1

1 1 1

2 1 2

3 2 1

4 3 2

we have $R_1 \bowtie R_2$

which has all tuples but 2

Extra tuples

$\therefore$ Not lossless.

Repeating not need to waste again

Note

For lossless join

- No Extra / spurious tuple generated at time of join
- No less tuple generated at join
- original tuple as it is generated after join from subrelation

Note

Lossy join

- $R_1 \cap R_2 = \emptyset$ then R Decomposition is lossy
no need to check

Unique $\rightarrow A'$ SK & Present at both sub relation
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Ex $\Rightarrow R(A, B, C)$

1	1	1
2	1	2
3	2	1
4	2	2

$R_1(A, B)$	$R_2(A, C)$
1 1	1 1
2 1	2 2
3 2	3 1
4 2	4 2

$R_1 = SK$

Soln $\Rightarrow$

$R_1 \bowtie R_2$

A	B	C
1	1	1
2	1	2
3	2	1
4	2	2

$\text{num(Attributes)} = 3$

$\text{num(tuples)} = 4$

$\therefore$ NO Extra/spurious tuple generated

Because of SK

$\downarrow$
 NO Redundancy $\therefore$ Decomposition is lossless

$\therefore$ NO less tuple missing/less

Note $\Rightarrow$ If any superkey present in all sub-Relation then the decomposition is always lossless, possibly

because the superkey will reduce redundancy & provide unique tuples.

$\rightarrow$ eliminate the possibility of generating of spurious tuples.

⇒ lossless join properties must satisfy

① attribute (R_1) $\cup$ attribute (R_2) = R (attribute)

All FD's of subRelation $\equiv$ All FD's of Base Relation

② $Attr(R_1) \cap Attr(R_2) = \emptyset$ then lossy decomposition

③ $[Attr(R_1) \cap Attr(R_2)] \Rightarrow Attr(R_2)$

or

$[Attr(R_1) \cap Attr(R_2)] \Rightarrow Attr(R_1)$

④ $sk \text{ in } R_1 \rightarrow sk \text{ in } R_2$

⑤ If any sk present in subrelation, then

there is possibility of lossless join.

Ex

$R(A, B, C)$

1	1	3
2	1	3
3	2	6
4	3	7

$R_1(A, B)$

1	1
2	1
3	2
4	3

$R_2(B, C)$

1	3
1	3
2	6
3	7

Repeating values
FD
Preserving,
Exists

Soln $\Rightarrow$

$A \rightarrow B \checkmark$
 $B \rightarrow A \times$

$B \rightarrow C \checkmark$
 $C \rightarrow B \checkmark$

$R_1(A, B)$ has
SK

$R_2(B, C)$ has SK then
 $\therefore$ there exists lossless
join

No need to check by
Natural join

$R_1 \bowtie R_2$

A	B	C
1	1	3
2	1	3
3	2	6
4	3	7

} same original
Relation $R \checkmark$

$\therefore$ lossless join exists

Note $\Rightarrow$ If in subrelations, any of the Attribute
is missing then entire join is lossy,
as the missing Attribute never be preserve

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$R(AB C D E)$

1	1	2	1	3
2	2	2	1	3
3	1	6	3	6
4	2	8	5	7
5	3	9	5	7

Q. 11

check $R_1(ABC)$, $R_2(DE)$

Soln

$R_1(ABC)$

↓

Contain A as SK

$A \rightarrow BC$

$R_2(D, E)$

↓

contain $D \rightarrow E$

'D' as SK

It means Nothing
common
between R_1 & R_2

But $R_1 \cap R_2 = \emptyset$ { violates lossless join }

for $\therefore$ lossy join

there must be a common
Attribute
in R_1 & R_2

(2) Decomposition (ABC, CD) check join is lossless or not?

Soln $R_1(ABC)$

$R_2(CD)$

↓

$A \rightarrow BC$

$A = SK$

↓

$C \rightarrow D$

$C = SK$

$\therefore$ ~~Decomposition~~ join is lossless ^{not}

But 'E' is not
Present

as Both subrelation has SK.

But one Attribute is

$\therefore$ lossy join

missing

paper Ex (4)

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check

(3) $R_1(A, B)$, $R_2(B, C, D, E)$

Soln $\rightarrow$ common Attribute = B

$B \neq SK$, $BC \neq SK$

$A \rightarrow B$, $B \rightarrow A$

$BC \rightarrow DE$ ($B \rightarrow C$ & $C \rightarrow E$)

~~$A = SK$~~

~~$BC = SK$~~

$\therefore$ lossy join

no SK

only $R_1(A, B)$ has SK

But $R_2(B, C, D, E)$ No single
Attributed SK

$\therefore$ lossy join

(4) check $R_1(A, B)$, $R_2(C, D)$, $R_3(D, E)$

Soln $\rightarrow$

$R_1(A, B) \bowtie R_2(C, D)$

no common Attribute

But now Can't say Anything
so check other
subrelation

$R_2(C, D)$, $R_3(D, E)$

$C \rightarrow D$

$D \rightarrow E$

$E \rightarrow D$

SK

SK

$\therefore$ lossy for this
combin

now check, $R_1(A, B) \bowtie R_{23}(CDE)$

$R_{23}(CDE)$

$\downarrow$

no common
Attributes

$\therefore$ Complete Relatⁿ is lossy

Ex. 2 → 10

 $R(A, B, C, D, E)$

1 1 2 1 3

2 2 2 1 3

3 1 6 3 6

4 2 8 5 7

5 3 9 5 7

① check $R_1(ABC)$ $R_2(CDE)$

Soln

↓	↓
$A \rightarrow BC$	$C \rightarrow DE$
$A = SK$	$C = SK$

∴ lossless join Both subrelation
has SK

∴ C is Common Attribute,

∴ All Attributes Present,

∴ Because of SK, no Extra spurious
tuples generated.

∴ lossless join

② check $R_1(ABC)$ $R_2(ABDE)$

Soln

↓	↓
$A \rightarrow BC$	$A \rightarrow BDE$

A is common in both subrelation &
is SK

∴ lossless join exists.

if Common Attribute is sk/ck then Relation is lossless by rule

Ex: $R(ABC D)$ check for lossless/lossy joins with FD sets.

FD: $\{A \rightarrow B, B \rightarrow C, C \rightarrow A\}$

$R_1(ABC)$

$R_2(CD)$

Soln: $\downarrow$

$A^+ = \{A, B, C\}$
 $C^+ = \{C, A, B\}$
 $C \rightarrow AB$ & $A \rightarrow BC$
 $A \neq sk$

$\left. \begin{array}{l} A \rightarrow B \\ B \rightarrow C \\ C \rightarrow A \end{array} \right\}$

$C^+ = \{C, A, B\}$
 C not derive any FD
 $\rightarrow$ Common Attribute 'C'

$R_1 \bowtie R_2$ (give all Attributes)
 $\{A B C D\}$

$\therefore$ ~~C is sk in R_1 , there is no need to check R_2 as it is 'C' common Attribute~~

This $\therefore$ Relation = lossless

$\rightarrow$ ① C is common Attribute in both R_1 & R_2

$\therefore R(ABC, D)$ Retrieved.

$\rightarrow$ ② $FD_1 R_1 \cup FD_2 R_2 \equiv$ Base Relation FDs

$\{A \rightarrow B, B \rightarrow C, C \rightarrow A\} \equiv \{A \rightarrow B, B \rightarrow C, C \rightarrow A\}$

$\therefore$ FD preserving

$\therefore$ Relation is lossless

→ solve the Right hand side only

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"Canonical cover / minimal cover / irreducible set of FDs"

for Any FD,

is FD (Canonical form),

if FD don't have,

- ① Extraneous Attribute
- ② Redundant FD

⇒ How to find minimal cover?

Ans: (Step-I) Splitting rule, so that in every FD right hand side has single attribute.

(Step-II) Remove Extraneous attribute.

(Step-III) Remove Redundant FD.

Ex ⇒

① ~~$AB \rightarrow C$ & $A \rightarrow C$~~ then Remove 'C' as it is Extraneous attribute

② $AB \rightarrow C$ & $A \rightarrow B$ Remove 'B' as it is Extraneous attribute

③ ~~$AB \rightarrow C$ & $B \rightarrow C$~~ Remove A

④ ~~$AB \rightarrow C$ & $B \rightarrow C$~~ Remove A

Canonical cover may not be unique

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Ex $\Rightarrow \{ AB \rightarrow C, C \rightarrow AB, B \rightarrow C, ABC \rightarrow AC, A \rightarrow C, AC \rightarrow B \}$

Solⁿ: $\Rightarrow$ split

Step-I Remove Attribute so right side must have single attribute,

$AB \rightarrow C, C \rightarrow A, B \rightarrow C, ABC \rightarrow A, AC \rightarrow B$
 $C \rightarrow B, ABC \rightarrow C, A \rightarrow C$
 $AC \rightarrow B$

Step-II

Remove Extraneous Attributes (remove Trivial FD)

$AB \rightarrow C, C \rightarrow A, C \rightarrow B, B \rightarrow C, ABC \rightarrow A, ABC \rightarrow C, A \rightarrow C, AC \rightarrow B$

$\{ \underline{B \rightarrow C}, C \rightarrow A, C \rightarrow B, \underline{B \rightarrow C}, A \rightarrow C, A \rightarrow B \}$

Step-III

Remove Redundant FDs.

$\{ C \rightarrow A, C \rightarrow B, B \rightarrow C, A \rightarrow C, A \rightarrow B \}$

Find ~~closure~~ closure of leftside which are derived all Attribute of next FD. Remove it.

$\rightarrow$ pick FD & try to find its closure exclude it if it closure possible then remove it.

$\{ C \rightarrow A, B \rightarrow C, A \rightarrow B \}$ $\checkmark$ minimal cover

Ex: $\Rightarrow R\{A, B, C, D, E\}$ check for relation is lossy or not

FD: $\{ AB \rightarrow C, C \rightarrow D, D \rightarrow EF, F \rightarrow A, D \rightarrow B \}$

Decomposition = $\{ ABC, CDE, EF \}$

Soln $\Rightarrow (ABC) \bowtie (CDE)$

AB C DE

A B C

$\downarrow$

C D E

$\downarrow$

$\checkmark C^+ = \{C, D, E, ABC\} = \text{sk}$

~~C $\rightarrow$ AB~~ $C \rightarrow DE$ $\checkmark$

$D^+ = \{D, B, E, F, A, C\} = \text{sk}$

~~AT~~ $A^+ = \{A, B, C\} = \text{sk}$

$B^+ = \{B\}$

$\checkmark C^+ = \{C, D, E, F, A, B\} = \text{sk}$

$C \rightarrow AB$ $\checkmark$

$E^+ = \{E\}$

C is sk in both subrelation

$\therefore R_1 \& R_2$

$(ABC) \& (CDE) = \text{lossless}$

Now, check for,

AB C D E

$\downarrow$
 $E^+ = \{E\}$

$\downarrow$
 $E^+ = \{E\}$

$F^+ = \{F, A\}$

$F^+ = \{F, A\}$

Here, no sk/sk
as E & F
both not derive
complete closure
set

is common
attribute,

subrelation ABCDE has sk, $\therefore$

All FDs will derived by ABCDE, But

No ~~union~~ $(ABCDE) \bowtie (EF) = \text{Empty}$ by the FD set
 $\therefore$ ~~lossless~~ join exists. as a lossy join.

Note

If 2 ^{sub} relations, of Relation R

R_1 & R_2 , Both have common Attribute

R_1 has sk , R_2 doesn't have ^{common Attribute} ~~any~~ sk / ck

$$R \Rightarrow FD = \{ \}$$

$\therefore$ Relation R = lossy decomposition

not lossless Relation.

decomposition is

Ex $\Rightarrow R(A, B, C, D, E)$ Check for Dependency Preserving?

FD: $\{ AB \rightarrow C, C \rightarrow D, D \rightarrow EF, E \rightarrow A, D \rightarrow B \}$

Decomposition (ABC, CDE, EF)

"Convert Relation from 1NF to 2NF"

Ex $\Rightarrow R(A, B, C, D)$

FD: $\{A \rightarrow B, B \rightarrow C\}$

Soln $\Rightarrow$ first check for highest NF

$$(ABCD)^+ = \{A, B, C, D\} = \text{sk}$$

$AB \neq D$

$$(AD)^+ = \{A, B, C, D\} = \text{sk}$$

ck = AD $\{ \text{As } A \ \& \ D \text{ not present in Right side} \}$

Prime Attribute = $\{A, D\}$

Non Prime Attribute = $\{B, C\}$

check for BCNF $A \rightarrow B \{ \text{BCNF } A = \text{sk} \}$
 $B \rightarrow C \{ \text{BCNF } \{ B = \text{not key} \} \text{ violate BCNF} \}$

check for 3NF, $B \rightarrow C \{ \text{NPA} \rightarrow \text{NPA} \therefore \text{violating 3NF} \}$

check for 2NF,

$A \rightarrow B$
 $\uparrow \quad \uparrow$
 part of key NPA

$\{ \text{violating 2NF, } \therefore \text{PD exists} \}$

$\therefore$ given Relation is in ~~3NF~~ 1NF

Now, decompose the $R(A, B, C, D)$

$R_1(A, B, C)$

$R_2(A, D)$

Keep common attribute which = ck

$A^+ = \{A, B, C\}$

$A^+ = \{A, B, C\}$

$A \rightarrow BC$

$D^+ = \{D\}$

$A = ck$

$FD_2 = \{ \}$

↑
single attribute
ck. $\therefore$ 2NF

$\therefore R_2$ is in BCNF

Exists,

No one to violate

Partial dependency
Property.

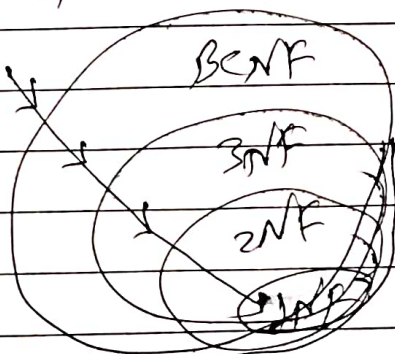
If any relation does not
generate any new FD
then it is in BCNF.

$\therefore R_1 \cup R_2 = 2NF$

$2NF \cup BCNF$

↑
Take the
highest one.

checking for
highest NF



$\therefore$ Relation R is in 2NF

Ex 1 R (A, B, C, D)

Convert 1NF to 2NF.

FD: $\{A \rightarrow B, C \rightarrow D\}$

Solⁿ $\Rightarrow$

Decomposed

R_1
ABC

R_2
ACD

Non Prime Attribute = $\{C\}$	$A^+ = \{A, B\}$	$C^+ = \{C, D\}$
Prime Attribute = $\{A, B, C, D\}$	$B^+ = \{B\}$	
	$C^+ = \{C, D\}$	$AC^+ = \{A, C, B, D\}$
$FD_1 = AC \rightarrow BD$	$AC^+ = \{A, B, C, D\}$	$FD_2 = AC \rightarrow BD$
	$AC = CK$	$AC = CK$

$\therefore$ By splitting property we get all FD of Relation

AC is common Attribute

And AC is CK in both sub relation,

$\therefore$ lossless decomposition & dependency preservation

$AC \rightarrow BD$

no Non Prime Attribute

$\therefore$ Never violate 2NF condition & no PD exists.

$\therefore$ Relation is in 2NF converted.

2NF $\rightarrow$ Contain Transitivity dependency

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"Convert Relation from 2NF to 3NF"

Ex $\Rightarrow$ $R(A, B, C, D)$

FD: $\{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$

Soln $\Rightarrow$

$(ABCD)^+ = \{A, B, C, D\} = SK$

$AB \not\rightarrow D$

$\downarrow$

$A^+ = \{A, B, C, D\} = SK$

$A = CK$, A is not in Right side
 $\therefore A$ is only CK.

Prime Attribute = $\{A\}$

non Prime

Attribute = $\{B, C, D\}$

$\rightarrow$ check for ~~BCNF~~ BCNF,

$A \rightarrow B$ { BCNF, $A = \text{key}$ }

$B \rightarrow C$ $\times$ violate BCNF

$\rightarrow$ check for 3NF

$B \rightarrow C$ { NPA $\rightarrow$ NPA $\therefore$ violate 3NF }

$\rightarrow$ check for 2NF

$A \rightarrow B$ { No subset of A, $C \rightarrow D$ { No PD } $\therefore$ Relation is in 2NF

$$FD = FD_1 \cup FD_4 \cup FD_5$$

$$= \{A \rightarrow B, B \rightarrow C, C \rightarrow D\} \therefore \underline{3NF} \text{ (All are in BCNF as well)}$$

$$\therefore R_1, R_2, R_3 \text{ (AB), (BC), (CD)}$$

are decomposition of for 3NF

cause lossy $\{A \rightarrow B, B \rightarrow C\}$ not derived

$$R_3 \{BCD\}$$

$$\downarrow R_4 \{BC\} \quad R_5 \{CD\}$$

$$FD_4 \{B \rightarrow C\}$$

$$\{C \rightarrow D\} FD_5$$

R_4 & R_5 are in 2NF

$FD_2 \{B \rightarrow C, C \rightarrow D\}$ having problems for 3NF,

$\therefore$ Decomposed Relation into R_1 & R_2
(ABC) (ACD)

$$R_1 (ABC)$$

$\downarrow$

$$A = CK, A^+ = \{A, B, C, D\}$$

$$A \rightarrow B$$

$$R_2 (ACD)$$

$\downarrow$

$$A = CK, A^+ = \{A, B, C, D\}$$

$$A \rightarrow CD$$

$$R_3 (BCD)$$

$$B \rightarrow C$$

$$B^+ = \{B, C, D\}$$

check for 3NF

$$\{A \rightarrow B\}$$

$$A \rightarrow CD$$

{Not violating 3NF}

A

$$B \rightarrow C$$

$$A \rightarrow C, A \rightarrow D$$

$(R_1 \& R_3) \rightarrow 2NF$ decomposition

Convert 3NF to BCNF, sometime

dependency preserving not

possible, sometimes

upto 3NF $\left\{ \begin{array}{l} \text{lossless} \\ \text{Dependency} \\ \text{Preserving} \end{array} \right\}$ possible.